به نام خدا

علی سپهرنیا (۹۶۱۲۱۰۳۸) | آزمایشگاه سیستم دیجیتال ۲ | استاد: خانم رضایی

پاسخ تمرین های جلسه هفتم

تمرین ۱:

برنامه ای بنویسید که یک شکل موج با فرکانس 1 KHz روی پایه OC0 ایجاد کند. (فرکانس داخلی میکرو را 1 MHz در نظر بگیرید).

G:\Personal\Uni\Micro Lab\Micro\Az7-Tamrin1\Az7-Tamrin1.c

Notes | Az7-Tamrin1.c

```
/*****************************************************
This program was created by the
CodeWizardAVR V3.12 Advanced
Automatic Program Generator
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Project : Session7-Exercise1
Version :
Date    : 11/10/2020
Author  : Ali Sepehrnia (96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>

// Declare your global variables here

void main(void)
{
// Declare your local variables here

// Input/Output Ports initialization
// Port A initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
```

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Notes | Az7-Tamrin1.c

```
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);

// Port B initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=Out Bit2=In Bit1=In Bit0=In
DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (1<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=0 Bit2=T Bit1=T Bit0=T
PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);

// Port C initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRC=(0<<DDC7) | (0<<DDC6) | (0<<DDC5) | (0<<DDC4) | (0<<DDC3) | (0<<DDC2) | (0<<DDC1) | (0<<DDC0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);

// Port D initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);

/*
OCR0=((((FREQ/2)/PRECSALER)/F_OUT)-1)
FREQ=1MHz
PRECSALER=256
F_OUT=1KHz
OCR0=1
*/
// Timer/Counter 0 initialization
// Clock source: System Clock
// Clock value: 3.906 kHz
// Mode: CTC top=OCR0
// OC0 output: Toggle on compare match
// Timer Period: 0.512 ms
// Output Pulse(s):
```

G:\Personal\Uni\Micro Lab\Micro\Az7-Tamrin1\Az7-Tamrin1.c

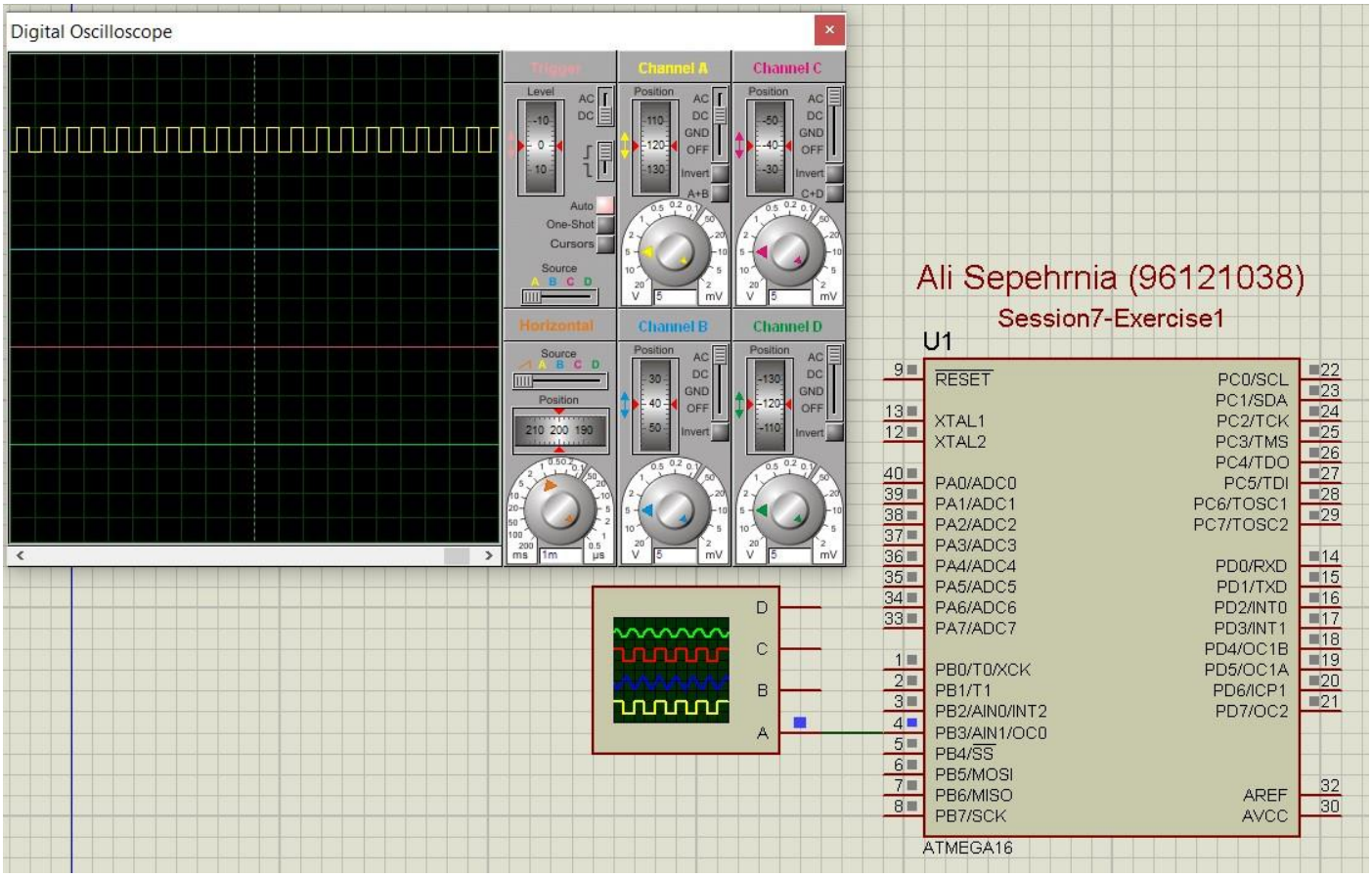
Notes | Az7-Tamrin1.c

```c
/*
OCR0=((((FREQ/2)/PRECSALER)/F_OUT)-1)
FREQ=1MHz
PRECSALER=256
F_OUT=1KHz
OCR0=1
*/
// Timer/Counter 0 initialization
// Clock source: System Clock
// Clock value: 3.906 kHz
// Mode: CTC top=OCR0
// OC0 output: Toggle on compare match
// Timer Period: 0.512 ms
// Output Pulse(s):
// OC0 Period: 1.024 ms Width: 0.512 ms
TCCR0=(0<<WGM00) | (0<<COM01) | (1<<COM00) | (1<<WGM01) | (1<<CS02) | (0<<CS01) | (0<<CS00);
TCNT0=0x00;
OCR0=0x01;

// Timer(s)/Counter(s) Interrupt(s) initialization
TIMSK=(0<<OCIE2) | (0<<TOIE2) | (0<<TICIE1) | (0<<OCIE1A) | (0<<OCIE1B) | (0<<TOIE1) | (0<<OCIE0) | (0<<TOIE0);

while (1)
      {
      // Place your code here

      }
}
```



تمرین ۲:

برنامه ای بنویسید که یک موج PWM با دیوتی سایکل ٪۲۵ تولید کند.

G:\Personal\Uni\Micro Lab\Micro\Az7-Tamrin2\Az7-Tamrin2.c

Notes | Az7-Tamrin2.c

```
/*****************************************************
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Project : Session7-Exercise2
Version :
Date    : 11/10/2020
Author  : Ali Sepehrnia (96121038)
Company :
Comments:


Chip type               : ATmega16
Program type            : Application
AVR Core Clock frequency: 1.000000 MHz
Memory model            : Small
External RAM size       : 0
Data Stack size         : 256
*****************************************************/

#include <mega16.h>

// Declare your global variables here

void main(void)
{
// Declare your local variables here

// Input/Output Ports initialization
// Port A initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRA=(0<<DDA7) | (0<<DDA6) | (0<<DDA5) | (0<<DDA4) | (0<<DDA3) | (0<<DDA2) | (0<<DDA1) | (0<<DDA0);
```

G:\Personal\Uni\Micro Lab\Micro\Az7-Tamrin2\Az7-Tamrin2.c

Notes | Az7-Tamrin2.c

```
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTA=(0<<PORTA7) | (0<<PORTA6) | (0<<PORTA5) | (0<<PORTA4) | (0<<PORTA3) | (0<<PORTA2) | (0<<PORTA1) | (0<<PORTA0);

// Port B initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=Out Bit2=In Bit1=In Bit0=In
DDRB=(0<<DDB7) | (0<<DDB6) | (0<<DDB5) | (0<<DDB4) | (1<<DDB3) | (0<<DDB2) | (0<<DDB1) | (0<<DDB0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=0 Bit2=T Bit1=T Bit0=T
PORTB=(0<<PORTB7) | (0<<PORTB6) | (0<<PORTB5) | (0<<PORTB4) | (0<<PORTB3) | (0<<PORTB2) | (0<<PORTB1) | (0<<PORTB0);

// Port C initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRC=(0<<DDC7) | (0<<DDC6) | (0<<DDC5) | (0<<DDC4) | (0<<DDC3) | (0<<DDC2) | (0<<DDC1) | (0<<DDC0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTC=(0<<PORTC7) | (0<<PORTC6) | (0<<PORTC5) | (0<<PORTC4) | (0<<PORTC3) | (0<<PORTC2) | (0<<PORTC1) | (0<<PORTC0);

// Port D initialization
// Function: Bit7=In Bit6=In Bit5=In Bit4=In Bit3=In Bit2=In Bit1=In Bit0=In
DDRD=(0<<DDD7) | (0<<DDD6) | (0<<DDD5) | (0<<DDD4) | (0<<DDD3) | (0<<DDD2) | (0<<DDD1) | (0<<DDD0);
// State: Bit7=T Bit6=T Bit5=T Bit4=T Bit3=T Bit2=T Bit1=T Bit0=T
PORTD=(0<<PORTD7) | (0<<PORTD6) | (0<<PORTD5) | (0<<PORTD4) | (0<<PORTD3) | (0<<PORTD2) | (0<<PORTD1) | (0<<PORTD0);

// Timer/Counter 0 initialization
// Clock source: System Clock
// Clock value: 1000.000 kHz
// Mode: Fast PWM top=0xFF
// OC0 output: Non-Inverted PWM
// Timer Period: 0.256 ms
// Output Pulse(s):
// OC0 Period: 0.256 ms Width: 0.064251 ms
TCCR0=(1<<WGM00) | (1<<COM01) | (0<<COM00) | (1<<WGM01) | (0<<CS02) | (0<<CS01) | (1<<CS00);
TCNT0=0x00;
OCR0=0x40; //64/256=0.25

// Timer(s)/Counter(s) Interrupt(s) initialization
TIMSK=(0<<OCIE2) | (0<<TOIE2) | (0<<TICIE1) | (0<<OCIE1A) | (0<<OCIE1B) | (0<<TOIE1) | (0<<OCIE0) | (0<<TOIE0);
```

G:\Personal\Uni\Micro Lab\Micro\Az7-Tamrin2\Az7-Tamrin2.c

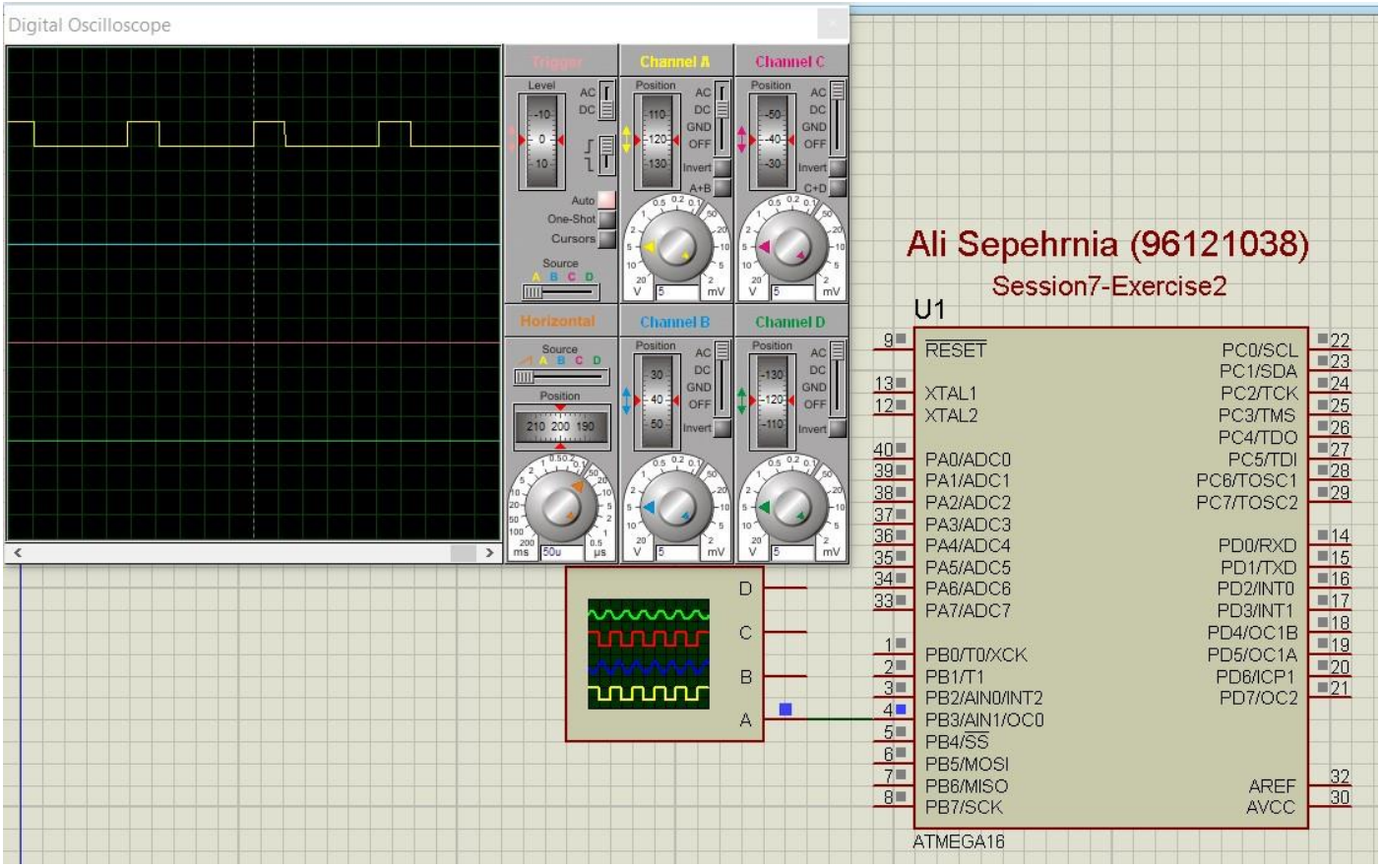
Notes | Az7-Tamrin2.c

```

while (1)
      {
      // Place your code here

      }
}

```


Digital Oscilloscope
Trigger
Channel A
Channel C
Horizontal
Channel B
Channel D
Ali Sepehrnia (96121038)
Session7-Exercise2
U1
RESET
XTAL1
XTAL2
PA0/ADC0
PA1/ADC1
PA2/ADC2
PA3/ADC3
PA4/ADC4
PA5/ADC5
PA6/ADC6
PA7/ADC7
PB0/T0/XCK
PB1/T1
PB2/AIN0/INT2
PB3/AIN1/OC0
PB4/SS
PB5/MOSI
PB6/MISO
PB7/SCK
PC0/SCL
PC1/SDA
PC2/TCK
PC3/TMS
PC4/TDO
PC5/TDI
PC6/TOSC1
PC7/TOSC2
PD0/RXD
PD1/TXD
PD2/INT0
PD3/INT1
PD4/OC1B
PD5/OC1A
PD6/ICP1
PD7/OC2
AREF
AVCC
ATMEGA16
D
C
B
A